

Chapter 44: Compositions and isomorphisms Worksheet

1. Are each of the following statements true or false?

- (a) Let P_n be the set of polynomials of degree at most n . Then P_n is isomorphic to $\mathbb{R}^n$.
- (b) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be an isomorphism, defined via $T(\mathbf{x}) = A\mathbf{x}$, for a A a 3×3 matrix. Then $\det(A) = 0$.
- (c) If $T : V \rightarrow W$ is a one-to-one linear transformation and $\dim(V) = \dim(W)$, then T is an isomorphism.
- (d) $T : M_{mn} \rightarrow M_{nm}$ by $T(A) = A^T$ is an isomorphism.
- (e) If V, W are two vector spaces where $\dim(V) = 4 = \dim(W)$, then every linear transformation $T : V \rightarrow W$ is an isomorphism.
- (f) It's possible to build a surjective linear transformation $T : \mathbb{R}^5 \rightarrow M_{32}$.
- (g) If $T : V \rightarrow W$ is a one-to-one linear transformation, and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set in V , then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ a linearly independent set in W .
- (h) If $T : V \rightarrow W$ is a linear transformation, then $\dim(\text{im}(T)) \leq \dim(V)$.
- (i) If $T : V \rightarrow W$ is a linear transformation that is onto, then $\dim(W) \leq \dim(V)$.
- (j) Suppose A is a 2×5 matrix with $\text{nullity}(A) = 3$, and $T : \mathbb{R}^5 \rightarrow \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, then T is an onto linear transformation.

2. Define $L : P_2 \rightarrow P_2$ by $L(p(x)) = x^2 p''(x)$.

Is L ...

- (a) Linear?
- (b) one-to-one?
- (c) onto?
- (d) An isomorphism?

3. Consider M_{22} , and consider the subset W of matrices A that are symmetric. For which d is W isomorphic to P_d ?

4. Consider $T : P_1 \rightarrow \mathbb{R}^2$ by $T(p(x)) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$.

Is T ...

- (a) Linear?
- (b) one-to-one?
- (c) onto?
- (d) An isomorphism?